Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/14/2005
PI Name:	Hammond, Timothy M.D.		
Project Title:	Transcription factors mediating gene expression changes during renal cell rotating wall vessel culture.		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	Alternate email: Timothy.Hammond@va.gov NOTE: PI previously at Tulane University until 2007 (Ed., 09/2010) Joint appointments Professor, Department of Internal Medicine, Nephrology Division, Duke University School of Medicine, and Staff Physician, Durham VA Health Care System.		
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	08/01/2001	**End Date:**	09/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Hammond, Timothy		
COI Name (Institution):			
Grant/Contract No.:	NAG9-1344		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Although there are several precedents for the transduction of mechanical forces into changes in gene expression, including heat shock proteins and shear stress responses, the molecular mechanisms by which mechanical culture conditions in the rotating wall vessel modulate gene expression in cultured cells remain largely unknown. Several characteristics of renal cells made they especially well suited for rotating wall vessel studies. First, reformation of microvilli, and expression of clinically important proteins such as P450 drug metabolism enzymes, and drug and hormone receptors provides a unique toxicology tool not available in any other form. Second, the in situ milieu of renal epithelial cells has shear levels similar to those in the rotating wall vessel. Last, several lines of evidence suggest defined transcription factors (from the vitamin D receptor, and TFII-I to TFII-B) translocated to the nucleus in a time dependent manner during initiation of rotating wall vessel culture. This application aims to utilize renal cells as a model system to identify and validate molecular mechanisms by which selected transcription factors control changes in specific clusters of genes during rotating wall vessel culture. Unifying hypothesis Nuclear translocation of specific transcription factors mediates changes in expression of specific gene clusters during rotating wall vessel culture of renal cells. Specific Aim I: To identify kinetic gene clusters responding to the “simulated microgravity” of rotating wall vessel mechanical culture conditions. Specific Aim II: To test whether blocking translocation of individual transcription factors specificity reverses the effects rotating wall vessel culture on select gene clusters. This aim will focus on the effects of up and down regulation of nuclear translocation of the vitamin D receptor nuclear translocation on gene expression clusters (Specific Aim IIa), as well as identifying other transcription factors for future studies (Specific Aim IIb). If successful, our work using micro-array technology will demonstrate specific clusters of genes, which change expression during the rotating wall vessel form of suspension culture, and some of the transcription factors, which mediate the changes. This will elucidate specific molecular mechanisms critical for optimization of suspension culture. This facilitates the production of bioproducts such as hormones from the kidney and other tissues, as well as new methods for testing drug interactions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work has two practical application to life on earth. The first application is to understand hoe to make cheap, clean bioproducts for use as biopharmaceuticals. This work is also designed to facilitate a quantum leap in studies of chemosensitivity by allowing individual drug regimens to be designed based on in vitro analysis of a patient diseased or normal tissue.
Task Progress:	Specific Aim I: To identify kinetic gene clusters responding to the “simulated microgravity” of rotating wall vessel mechanical culture conditions. The changes in gene expression during culture of human renal cells in the rotating wall vessel were compared to the changes in the kidney of mice flown in the true microgravity of space. A publication reporting these changes is in preparation. Specific Aim II: To test whether blocking translocation of individual transcription factors specificity reverses the effects rotating wall vessel culture on select gene clusters To determine the effects of the vitamin D receptor, a transcription factor identified as one of the factors moving to the nucleus in Specific Aim 1, we tested for the ability to inhibit this translocation with vitamin D receptor tail peptides. Inhibition of vitamin D receptor did not increase the production of urokinase, but blocking the urokinase receptor with neutralizing antisera did increase urokinase production. Specific Aim III [Flight]: To identify kinetic gene clusters and transcription factor translocation in response to the true microgravity of space flight mechanical culture conditions. This renal epithelial cell data has been compared for organ specificity using data sets from human liver cells flown in space by Stelsys LLC, to the whole organ kidney and liver effects by comparison to tissue gene expression in mice flown on STS-108 by Amgen, and mechanistically by comparison to a simple eukaryotic system in the yeast Saccharomyces cerevisiae. This has allowed comparison of organ specific changes, determination how good ground-based models are as predictors of microgravity effects, and prediction of transcription mediated drug metabolism changes in whole animals and astronauts in space.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Abstracts for Journals and Proceedings	Johanson K, Allen PL, Gonzalez-Villalobos R, Nesbitt J, Nickerson CA, zu Bentrup KH, Wilson JW, Ramamurthy R, D’Elia R, Muse KE , Freeman J, Stodieck LS, Hammond TG. "Haploid deletion strains of Saccharomyces cerevisiae that determine survival during space flight." 15th Humans in Space Symposium, Graz Austria, 22-26 May, 2005. Proceddings of the 15th Humans in Space Symposium, May 2005. , May-2005
Articles in Peer-reviewed Journals	D'Elia R, Allen PL, Johanson K, Nickerson CA, Hammond TG. "Homozygous diploid deletion strains of Saccharomyces cerevisiae that determine lag phase and dehydration tolerance." Appl Microbiol Biotechnol. 2005 Jun;67(6):816-26. Epub 2004 Dec 10. PMID: 15968571 , Jun-2005
Articles in Peer-reviewed Journals	Klaus DM, Benoit MR, Nelson ES, Hammond TG. "Extracellular mass transport considerations for space flight research concerning suspended and adherent in vitro cell cultures." J Gravit Physiol. 2004 Mar;11(1):17-27. PMID: 16145796 , Mar-2004
Patents	60/574,756. Full patent filed May 2005. May-2005 Hammond, Timothy G. and Patricia L Allen. "Production of urokinase by immortalized human renal epithelial cells on beads."